

MULTIMODAL RECOGNITION

Week 5: Diffusion models

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MASTER IN
COMPUTER VISION
Barcelona



GitHub Repository

https://github.com/JordiVentosa/C5_Team3

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1. Introduction

Previous Week

In the previous week, we worked on image captioning using **transformer-based architectures**. We adopted a progressive methodology to incrementally update our initial baseline.

This **baseline** originally consisted of a pretrained **ViT encoder** paired with a **GPT2 decoder**. Through systematic experimentation, we explored the impact of fine-tuning only the captioner, only the backbone, or all parameters simultaneously. Furthermore, we tested alternative encoder-decoder combinations and evaluated three zero-shot **Qwen vision-language models** under basic and advanced prompting strategies, as well as a custom **ViT+LoRA-adapted Qwen3.5** pipeline.

Our results demonstrated that the most significant performance improvement came from fine-tuning only the GPT2 captioner, which reached a **METEOR of 0.424** and gained the ability to identify unrecognizable images. For the LLM-based approaches, prompt engineering proved to be the key factor, with advanced prompting improving METEOR by up to +9.5 points at zero additional computational cost.

Ultimately, the experiments also highlighted that large pretrained multimodal models already have strong visual understanding, and that the bottleneck in zero-shot settings is often how the question is asked rather than model capacity.

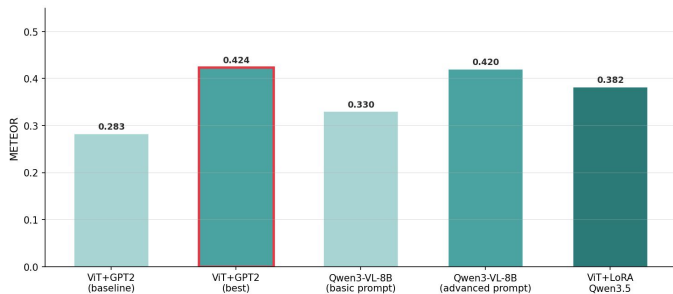


Best model Task 1
(ViT 🧠 & GPT2 🔥):

"a bottle of water sitting on a desk next to a computer keyboard and mouse"

Best model Task 2
(Qwen3.5-9B - Advanced prompt):

"a clear bottle of germ-x hand sanitizer with "dapacom" handwritten on the label, sitting on a desk next to a keyboard."



2. Qualitative comparison of Diffusion models

Stable diffusion 1

We wanted to visualize the evolution in quality between the different diffusion models, so we started with the initial versions of Stable Diffusion. The **LCM** model uses **Latent Consistency Models** to reduce the needed number of steps to just 4. As we can see, images aren't great, we can see a sort of frog on a bike, with the LCM model actually putting the bike in space.

SD 1.4



SD 1.5 LCM



SD 1.5



Stable diffusion 2

On this second version of Stable Diffusion, the encoder was moved to a **CLIP encoder**, leading to better understanding of the text prompt. We can see that now the frogs look more like frogs, and both images correctly place the motorcycle in space.

SD 2.1



SD 2.1 Turbo



Stable diffusion XL

The XL models show a really big jump from **SD 2.x**, the frogs look sharper and more realistic, there is more detail showing that we are in space, and the limbs of the frog appear correct. The **Turbo** model is distilled to run in just 1 step, hence the lower quality.

SD XL



SD XL Turbo



Stable diffusion 3.5

On **SD 3.5**, the architecture moves to a **Multimodal Diffusion Transformer** instead of the typical **UNet**, resulting in a significant improvement in quality. Within the same family, **Medium** already looks really good, but the jump to **Large** brings better sharpness, more lighting consistency, and overall more realistic output. The **Turbo** variant, similar to **XL-Turbo**, shows a sort of haze across the entire image.

SD 3.5-medium



SD 3.5-large



SD 3.5 -large-turbo



Flux

Flux moves away from diffusion and uses **rectified flow instead**. The results from **Flux.1** were already quite impressive, but **Flux.2** showed a further leap in image quality. The main issue we encountered with Flux.2 was its size, while we were able to run the full model on 2 H100 GPUs, we ended up using the **4-bit quantized version**, which fits on a single ADA 6000.

FLUX.1-dev



FLUX.1-schnell



FLUX.2-dev



Other

We also wanted to test a couple of models outside the Stable Diffusion and Flux families. It was nice to see how well **AuraFlow** handled the reflection of the bike, even though there probably should not be a reflection there.

playground-v2.5



auraflow-v0.2



3. Experiments on model inference

DDIM vs DDPM

DDIM, being a **deterministic** sampler, produces **usable** images even at just **10** steps, with the scene already coherent and the subject well defined. **DDPM**, being **stochastic**, **requires** significantly **more steps** to converge, at **10 steps** the output is largely abstract, with severe **color artifacts**, only becoming comparable to DDIM around 40 steps.

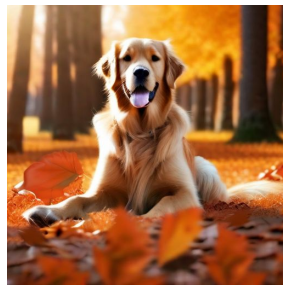
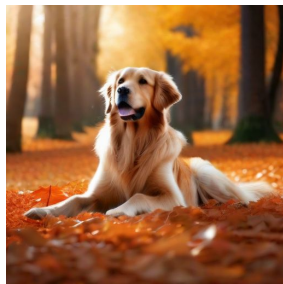
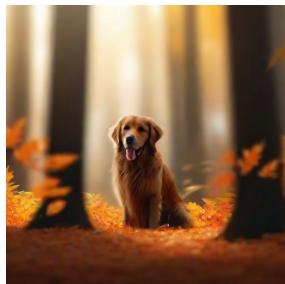
Steps

10

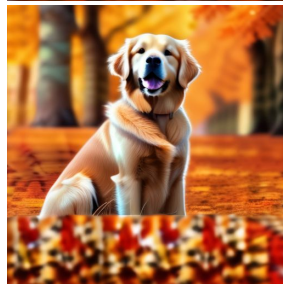
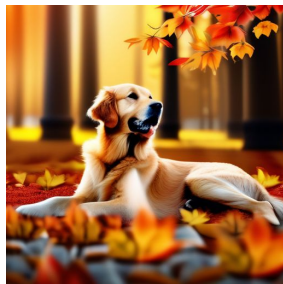
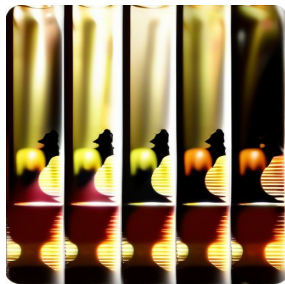
20

40

DDIM



DDPM

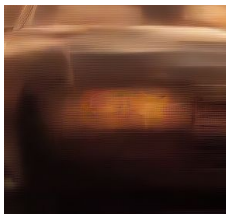
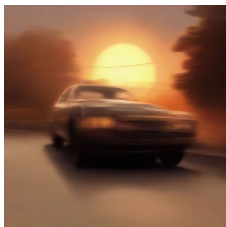


Number of steps

A good number of steps has proven key to generating good images. If the step count is **too low**, images appear **blurry**, the scene is **not accurate** to the prompt, and some **color artifacts** can appear. **Higher** is **not necessarily better** though, as we have seen that Stable Diffusion XL generates almost identical images from 30 steps onward.

Steps

5



10



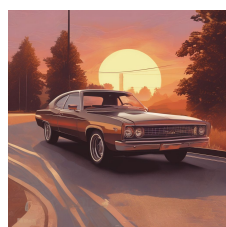
20



30



40



CFG

CFG is a parameter that **regulates how closely** the **model** follows the given **prompt**. Through testing we have seen that **lower CFG** values often lead to **black and white**, sketch-like images, and as CFG **increases** the model starts generating more **realistic** results, but also **over-saturates** the **colors** quite a lot, creating an overexposed look. We can also see that at low CFG values, elements explicitly mentioned in the prompt, such as the woman we asked for, do not appear at all.

CFG = 1



CFG = 3



CFG = 10



Negative and Positive prompting

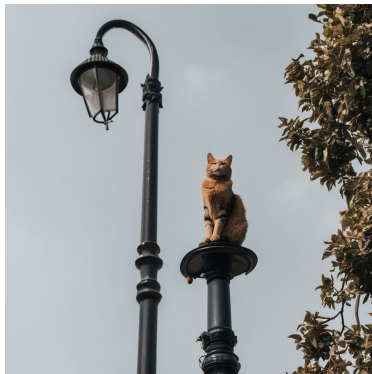
We wanted to test how **prompting different styles**, using both **positive** and **negative prompting**, would affect the generated images. We chose to prompt a cat sitting on top of a lamp. We were surprised by how much the **negative** prompt **affected** the **output**, yielding higher quality than even the positive prompt that explicitly asked for a high quality image.

Original



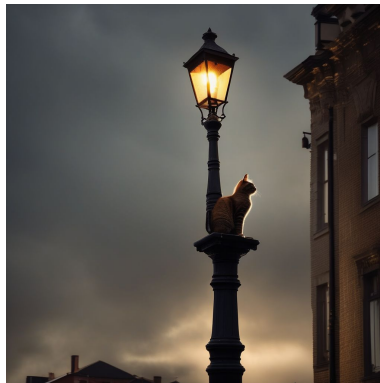
"A cat sitting on top of a lamppost"

Positive Prompt



Here we add a style prompt, asking for high quality, professional, cinematic.

Negative Prompt



Here we do a negative prompt for daytime, indoor, blurr and low quality.

4. Analysis of our Captioning Model

General Discuss and Recap

Last week we tested different models and obtained different metrics, as it can be observed in the table.

For this week, we chose to try to improve the **ViT+Qwen2.5-4B** model, since it has the **best metrics** in **BLEU-1**, **BLEU-2** and **ROUGE-L**. Although it shows a **lower METEOR** score than **ViT+GPT-2**, this is not necessarily a regression.

The ViT+GPT-2 model inflates its METEOR by relying on memorized templates and, by overusing the dataset's standard quality refusal phrase, which happens to overlap with ground truth stems. In contrast, Qwen attempts to describe the actual visual content rather than defaulting to safe probabilistic predictions, making it a stronger foundation for this task despite the metric gap.

That said, both models share a common failure: the quality fallback problem, and as we will see, Qwen suffers from it significantly more than GPT-2.

Last Week Results					
Task	Architecture	Meteor	Rouge-L	Bleu-1	Bleu-2
Task 1	Baseline	0.283	0.367	0.542	0.340
	Captioner	0.424	0.429	0.495	0.362
	Backbone	0.420	0.356	0.278	0.208
	All	0.414	0.357	0.270	0.202
Task 2	Zero Shot	0.415	0.405	0.499	0.334
	LoRA	0.382	0.441	0.572	0.423

Model Fail Cases: Quantitative

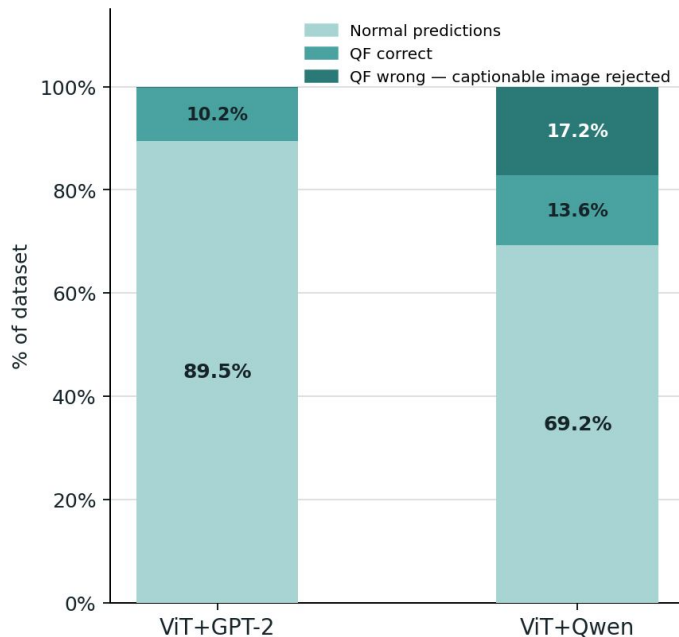
Both models frequently output a standard **refusal string** ("quality issues are too severe to recognize visual content") whenever the estimate an image is too degraded to be described. However, their reliability is very different.

The **ViT+GPT-2** setup applies this fallback to just **10.5%** of the images and gets it right **96.8%** of the time. Out of **7,750** samples, it only falsely rejects **26** images.

In the other hand, ViT+Qwen triggers the fallback on **30.8%** of images, but over half of these are false positives. This makes the model miss predict **1,336 perfectly** captionable images, which if a **17.2%** of the full dataset.

Because the Qwen shares no meaningful text with the ground truth in these cases, it performance drops. ROUGE-L drops to zero in over half of these false rejections, and METEOR collapses to near 0.001. We can conclude, its quality threshold is excessively aggressive, causing it to confidently reject images that human annotators had no trouble describing.

How often does each model fall back?



Model Fail Cases: Qualitative

The following examples show cases where **ViT+Qwen** outputs the **quality fallback** string despite the **image** being **perfectly describable**. In all cases, human annotators provided clear and consistent captions across all five references.

Predicted caption	Reference examples
"quality issues are too severe to recognize visual content"	"a computer screen shows a repair prompt" "partial computer screen showing the need of repairs"
"quality issues are too severe to recognize visual content"	"a dell laptop screen showing Windows 7 home premium" "a computer screen with a windows dialogue box"
"quality issues are too severe to recognize visual content"	"a tv screen displaying a news broadcast and a shelf with toys" "a desk with a television and a shelf above with toys"
"quality issues are too severe to recognize visual content"	"a hardcover novel from Clive Cussler and Dirk Cussler" "a book with an image of a sci-fi boat on the cover"
"quality issues are too severe to recognize visual content"	"a white orca membership card laying on a wooden table" "a white card laying on a wooden surface"

5. Synthetic Data Generation

Synthetic Data Generation Pipeline 1

Collect Caption Examples from VizWiz

1. Take all caption sets from the VizWiz training annotations
2. Select sets where at least one annotator reported quality issues, but no more than 2 did
3. From those sets, keep only the non-quality-issue captions to obtain:

Clean pool of VizWiz caption sets

Generate new captions with Qwen3.5-35B-A3B

Sample 5 random caption sets from the pool as few-shot examples and prompt the model to generate a new caption set containing:

1 global summary caption

5 VizWiz-like captions

Repeat until enough synthetic captions

FLUX.2 Prompt:

\$GLOBAL_SUMMARY_CAPTION + " IMAGE STYLE: Close taken smartphone photo with severe photographic problems: Severe motion blur or out of focus, subject is partially cut off, the photo is an accidental tilt, it appears a shaky camera, the lighting has hutter lag blur and is poorly framed."

Generate Image using FLUX.2-dev-bnb-4bit

Concatenate the global caption with a fixed style degradation prompt to replicate VizWiz photographic conditions and generate:

Synthetic Image

Repeat for each synthetic caption set

Qwen3.5-35B-A3B output example:

"global_caption":

"A blurry black remote control with white buttons rests on a beige couch.",

"captions":

"A dark remote sits on the sofa cushion.",

"The white buttons on the remote stand out against the beige couch.",

"A black remote lies on the cushion.",

"A blurry remote control rests on the beige couch.",

"A remote control rests on the sofa."

Synthetic data:

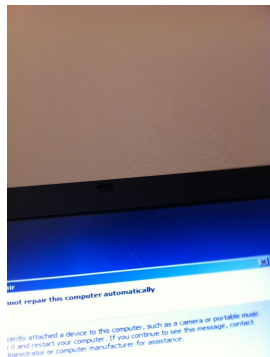
Degraded-quality images paired with 5 clean ground truth captions, following the VizWiz format

Synthetic Data Generation Pipeline 1

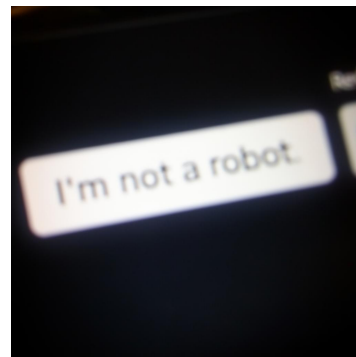
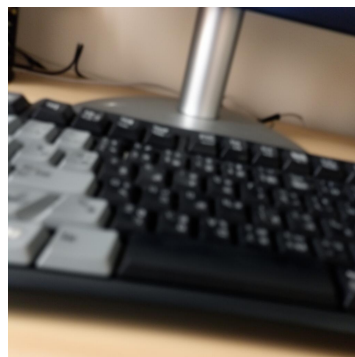
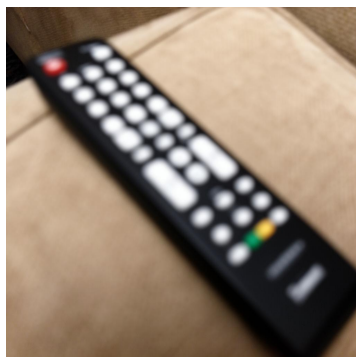
The VizWiz training set contains **23,954 images**. To complement it with additional training data, we **generated 2,000 synthetic images** using [FLUX.2-dev-bnb-4bit](#), a **4-bit quantized** version of **FLUX.2-dev** that we considered that generates good enough images at much lower memory cost than FLUX.2-dev. The generated images showed results similar enough to real VizWiz samples to be useful as training data, replicating the special characteristics of the dataset (motion blur, poor framing, accidental tilt and low lighting...).

Below are some examples, while the synthetic images are not identical to real VizWiz samples, they successfully capture the visual style and quality conditions of the dataset.

Real VizWiz



Synthetic
(FLUX.2-dev-bnb-4bit)

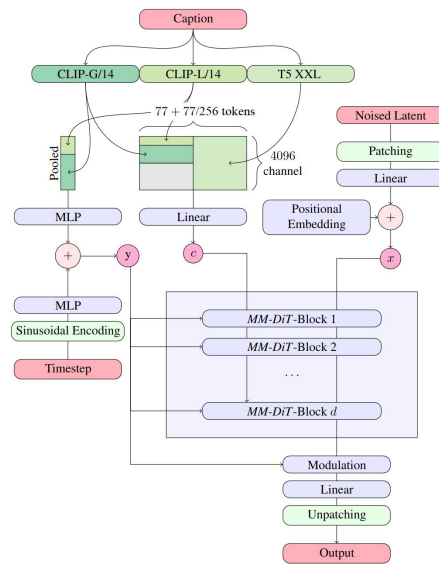


Extra experiment: Fine-tune Stable Diffusion 3.5

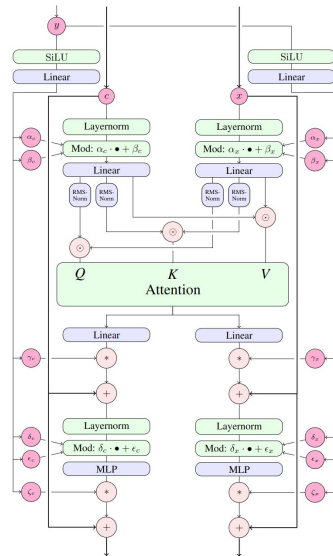
Since generating images close to our dataset distribution was difficult, we tried fine-tuning Stable Diffusion 3.5 Medium on the VizWiz training set images. Given the size of the model, we decided for a **LoRA** approach to keep the training feasible.

However, we faced different difficulties using the standard fine-tuning repositories, and we ended up using [kohya-ss/sd-scripts](https://github.com/kohya-ss/sd-scripts).

We performed different tests, but the most interesting one used **4,000 training steps** over **4 epochs** on the full training set. Despite this, the qualitative results shown in the next slide show that the model still struggles to generate images that are close to the VizWiz distribution.



(a) Overview of all components.



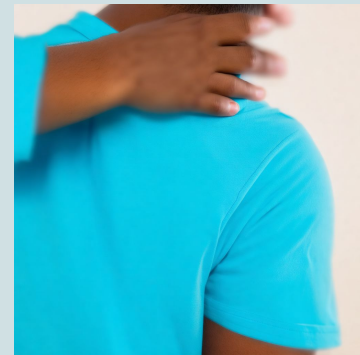
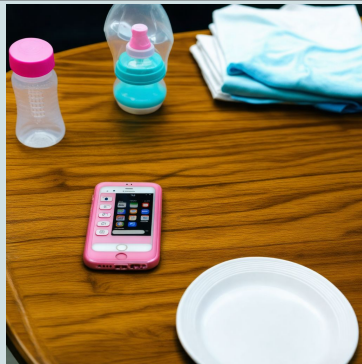
(b) One MM-DiT block

Stable Diffusion 3.5 architecture

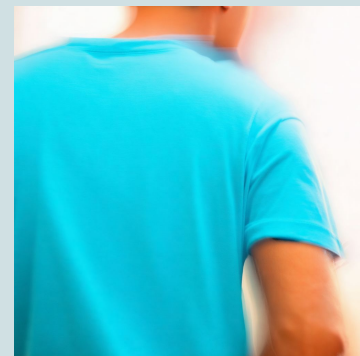
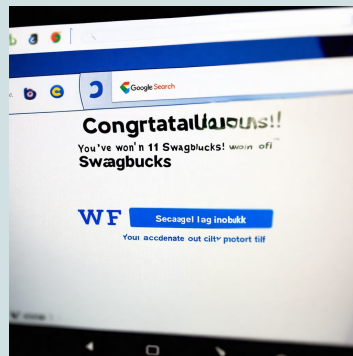
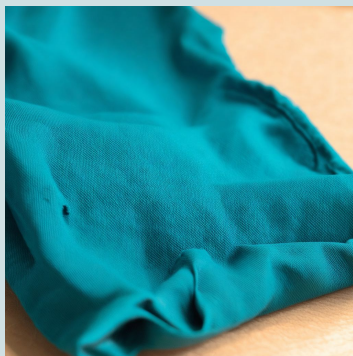
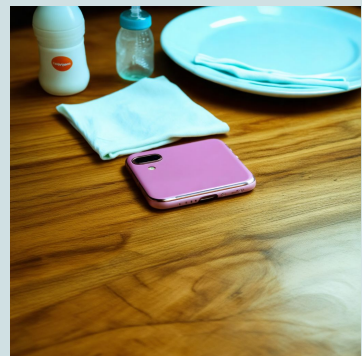
Extra experiment: Fine-tune Stable Diffusion 3.5

Stable Diffusion 3.5

(using same prompts)

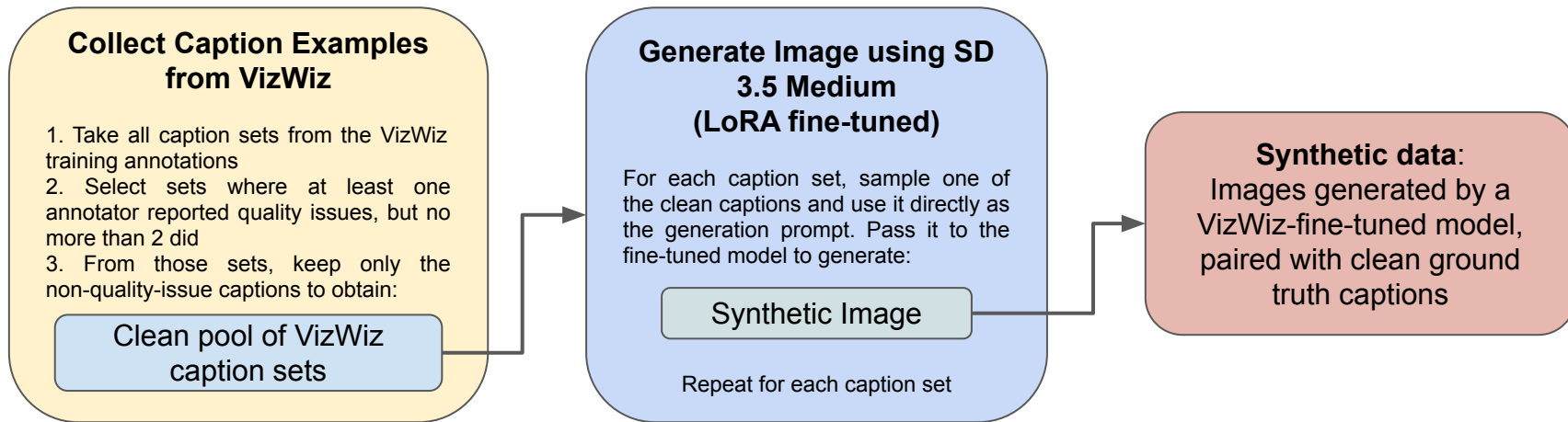


**Stable Diffusion 3.5
fine-tuned**



Synthetic Data Generation Pipeline 2

We decided to explore a different approach to synthetic data generation. Unlike the FLUX.2 pipeline, this one **uses real VizWiz captions** directly as generation prompts (skipping the Qwen caption generation step) to test whether the fine-tuned model alone could replicate the dataset distribution. Despite the fine-tuning not having achieved the results we hoped for, we decided to use it in this pipeline to generate **2,000** images and see whether using real VizWiz captions as prompts could compensate and bring the outputs closer to the dataset distribution.

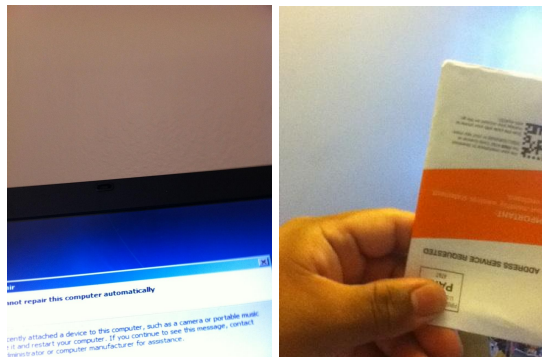


Synthetic Data Generation Pipeline 2

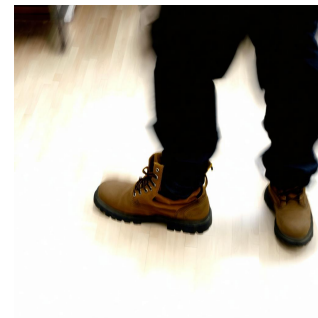
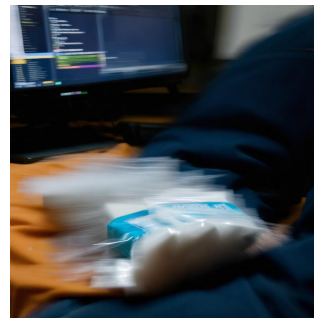
As we can see, **using** the real **VizWiz captions** as prompts **improved** the **results** compared to using the fine-tuned model with a generic prompt. The **generated images** are visually **reasonable**, but qualitatively they appear further from the VizWiz distribution than the FLUX.2 quantized outputs.

We can noticed an **excessive** use of **blur** across most generated images, which goes beyond the natural degradation seen in the real dataset. Additionally, some **outputs** are **unrealistic** (such as the image of a hole in the middle of a hallway floor) suggesting the fine-tuned model still struggles to stay within the dataset distribution.

Real VizWiz



Synthetic
(SD 3.5 Medium
fine-tuned)



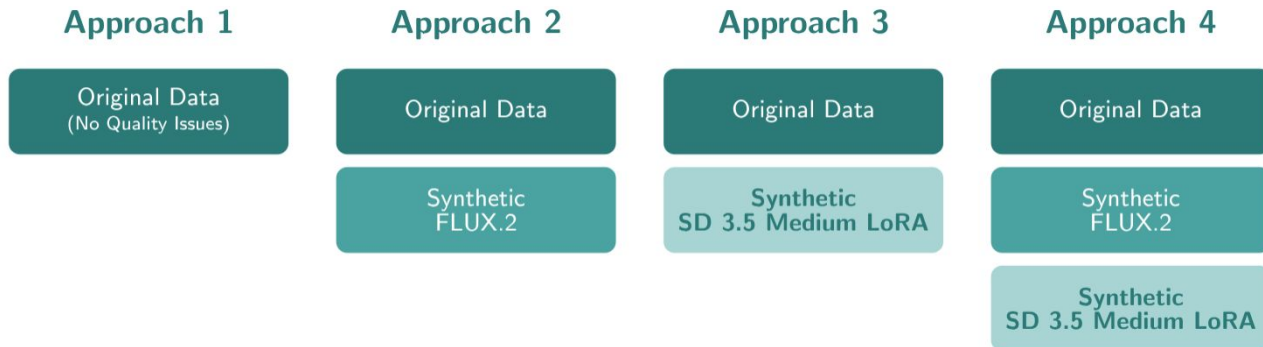
6. Finetuning with synthetic data

Training strategies

Through the previously described pipelines, we generated 2,000 images using FLUX.2-dev-bnb-4bit and 2,000 images using the LoRA fine-tuned SD 3.5 Medium. We used this synthetic data to retrain the ViT+Qwen3.5 captioning model under 4 different training strategies:

1. Using **only** the **original data**, but **removing quality issue** captions from images that had at least two valid captions
2. Using the **original data combined** with the **FLUX.2** synthetic images
3. Using the **original data combined** with the **SD 3.5 Medium LoRA** synthetic images
4. Using the **original data combined** with **both synthetic sets**

The idea for the first train was to see if data augmentation was really necessary or simply constraining the model to see less quality issue captions would be enough.



Quantitative Results

We were surprised by **how much removing quality issue captions negatively affected the scores**. It seems that forcing the model to avoid predicting that label, without providing enough alternative training examples, was too punishing.

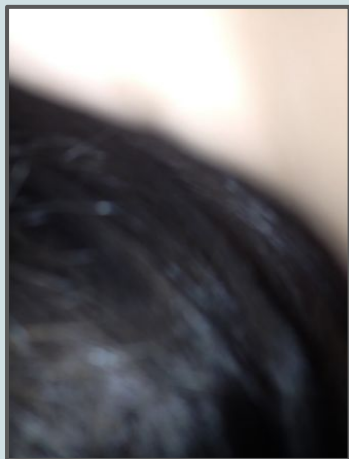
When **including synthetic data, quantitative results** also appear to have **decreased**, especially for the FLUX.2 with Qwen captions approach. We believe the **gap** between the Qwen captions approach and the **SD 3.5** one comes from the fact that **Qwen's captions did not reflect the dataset's style** closely enough, causing the model to learn a different caption syntax.

To fully understand whether the fine-tuned models actually improve over the baseline, we need to take a deeper **look** at the **qualitative results**.

Approach	Meteor	Rouge-L	Bleu-1	Bleu-2
Best Week 4	0.382	0.441	0.572	0.423
Original Data (No Quality Issues)	0.319	0.256	0.234	0.152
Original Data + Synthetic FLUX.2	0.381	0.417	0.551	0.366
Original Data + Synthetic SD 3.5 Medium LoRA	0.373	0.421	0.609	0.412
Original Data + Synthetic FLUX.2 + Synthetic SD 3.5 Medium LoRA	0.356	0.396	0.571	0.368

Qualitative results

In this example we can see a clear example of the model performing better, but being punished because no caption gave anything but quality issues. With this augmented approach we were able to identify an image that even the captioners did not label.



Reference: Quality issues are too severe to recognize visual content.

Removed captions: Quality issues are too severe to recognize visual content.

FLUX-2: A close-up view of someone's head with dark hair

SD-3.5: A black hair with a white background

FLUX-2 + SD-3.5: Quality issues are too severe to recognize visual content.

Qualitative results

In this other example, some captions also define the image as quality issues, but none of the models we trained took that route.



Reference: Quality issues are too severe to recognize visual content.

Removed captions: A computer screen with a blue background and white text

FLUX-2: A computer screen with a blue background and the windows start menu open

SD-3.5: A computer screen with a blue background and white text

FLUX-2 + SD-3.5: A computer screen with a blue background and the windows start menu open

7. Conclusions and Summary

General Discuss and Recap

Quantitatively, the models **learning to not predict quality issues** lead to **worse overall performance**, which shows that the dataset heavily relies on the quality issue caption whenever the image is not perfectly clear.

Qualitatively though, we can clearly see that this **label is overused**, and some images that were previously marked as quality issues are clear enough to generate meaningful captions.

We believe the **fine-tunes** were **successful**, because even though the metrics decreased, there are now a lot of images that were previously not labeled that the model can now accurately describe.

Approach	Meteor	Rouge-L	Bleu-1	Bleu-2
Best Week 4	0.382	0.441	0.572	0.423
Original Data (No Quality Issues)	0.319	0.256	0.234	0.152
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Original Data + Synthetic FLUX.2 + Synthetic SD 3.5 Medium LoRA	0.356	0.396	0.571	0.368

Summary Slide



Conclusions

Diffusion models

There is a huge variety of diffusion models, each offering fine-grained control over the generation.

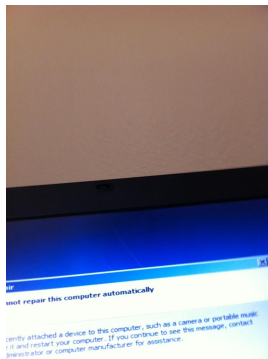
Analysis Captioning Model

ViT+Qwen2.5-4B from previous week is prone to predicting the quality fallback string instead of a caption.

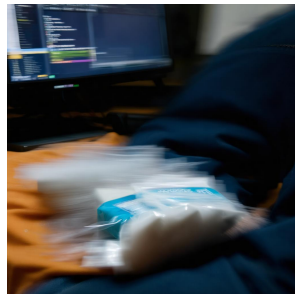


Qualitative:

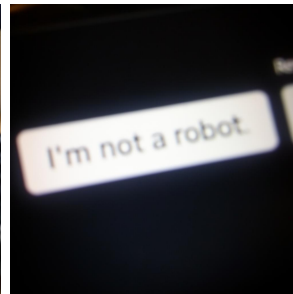
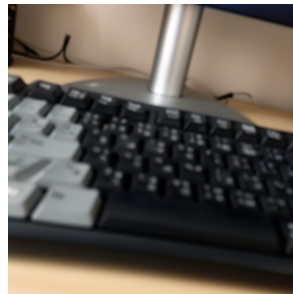
Real VizWiz



Synthetic
(SD 3.5 Medium fine-tuned)



Synthetic
(FLUX.2-dev-bnb-4bit)



Quantitative :

Approach	Meteor	Rouge-L	Bleu-1	Bleu-2
Best Week 4	0.382	0.441	0.572	0.423
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